

Jiaming Deng

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Work Experience

TikTok | Dublin, Ireland

Nov. 2023 – Present

Software Engineer II

Mar. 2025 – Present

Build distributed storage infrastructure for TikTok's video platform, focusing on object caching, edge storage, and cache addressing with **Go**, **Python**, and **Redis**.

- **Distributed Object Cache Optimization:** Improved large-scale video frame delivery by introducing **multi-level cache routing** across local cache, remote peer cache, **RDMA cache**, and origin fallback, with asynchronous write-back, capacity-aware eviction, and runtime feature gates to reduce origin load and improve cache availability.
- **Edge Storage Service:** Built a large-scale edge storage redundancy system for 12k+ edge video uploads qps, improving upload resilience through local object writes, dual-replica async replication, automatic peer discovery, WAL-based retry, and disk-health governance across multi-PiB edge infrastructure.
- **Decentralized cache addressing:** Designed and implemented a **consistent-hashing-based** cache addressing architecture with **virtual nodes** and **Gossip-based topology discovery**, replacing centralized metadata lookup with direct client-to-node routing to reduce lookup latency and improve read/write throughput.
- **On-call AI Agent:** Built a supervisor-based multi-agent system with specialized MCP tools for automated root cause analysis and self-recovery across **5 video platform products**; published reusable **Grafana MCP tools** to TikTok's internal marketplace.

Software Engineer I

Nov. 2023 – Mar. 2025

- **Object Cache Service:** Developed and maintained a high-throughput object cache service for video frame downloads, supporting local cache reads, origin fallback, asynchronous cache write-back, and cache prewarming, serving **10M+ QPS** with an **82%+ cache hit rate** across TikTok's video platform.
- **Centralized cache addressing service:** Developed and maintained a high-throughput cache addressing service backed by **in-memory indexing and Redis**, enabling reliable object-to-cache-node lookup for small-object video caching at large scale.

Trinity College Dublin – Gillan Lab

Jun. 2023 – Oct. 2023

Software Engineer - Research Assistant

Dublin, Ireland

- **Research web applications:** Developed and tested **React/JavaScript** web applications and **MongoDB**-backed services for Gillan Lab research staff to view, analyze, and export neuroscience research data, supporting evolving research requirements.
- **Data retrieval optimization:** Improved research data retrieval performance by **30%** by reducing redundant data processing and optimizing streaming pipelines and Gzip compression, supporting more efficient access to experimental data.
- **Participant data access control:** Built a permission-management web application and backend services that enabled researchers to manage access to participants' experimental data and enforce access permissions across research workflows.
- **Developer tooling and documentation:** Documented web server API endpoints and made them accessible through a web interface, helping developers inspect the services supporting the lab's research applications.

Education

Trinity College Dublin

Sep. 2022 – Jun. 2023

Master of Science in Computer Science

Dublin, Ireland

Xi'an Jiaotong University

Sep. 2018 – Jul. 2022

Bachelor of Engineering in Computer Science

Xi'an, China

Technical Skills

Languages: Go, Java, Python, JavaScript, C++

AI & Agent Tools: Eino, LangGraph, LangChain, MCP, Codex, Claude Code

Backend Frameworks: Gin, Spring Boot, Express, Kitex, Hertz

Data & Infrastructure: Redis, MySQL, MongoDB, RocketMQ, Object Storage, Kubernetes